

The Relationship Between Renewable Electricity Generation, Fossil Fuel Production, and Electricity Demand in the UK Power Grid (2019-2021)

25th March 2026

1 Overview

This project investigates the relationship between renewable electricity generation and fossil fuel electricity production in the UK power grid, and examines how electricity demand influences this relationship. Understanding these interactions is important as the UK continues to develop renewable energy sources while being less reliant on fossil fuels as part of its transition to a low-carbon energy system.

The analysis uses UK power grid data collected between 2019 and 2021. This period includes the COVID-19 pandemic, which provides context for potential changes in electricity demand and generation patterns. The dataset was processed in the Python language using pandas library, including cleaning column names, converting timestamps, and creating variables representing total renewable generation and total fossil fuel generation. Anomalous observations containing implausible values were identified and removed before analysis, whereas some were plausible given the context of the world systems.

Exploratory data analysis was conducted using summary statistics and visualisations to examine and analyse patterns and relationships within the data. Scatter plots, a correlation heatmap, and a timeseries plot were used to illustrate relationships between renewable generation, fossil fuel generation, and electricity demand. To model and measure these relationships, a multiple linear regression model

was fitted with fossil fuel generation as the dependent variable and renewable generation and electricity demand as predictors.

The results indicate a strong negative relationship between renewable generation and fossil fuel generation, while electricity demand shows a strong positive relationship with fossil fuel generation.

2 Introduction

Context and motivation This area of study covers the relationship between renewable energy and fossil fuel production and how electricity demand affects this. The electric power grid is a network that produces, transmits, and dispenses electricity to industries and households. Electricity is produced in power plants, where mechanical energy is converted into electrical energy. This can come from fossil fuels, nuclear energy, or renewable sources such as solar, wind, and hydropower. [1]

This is interesting to investigate because there has been a transition to using more renewable sources to combat climate change and create a sustainable future. Understanding how this relationship works is critical to seeing the way modern electricity and, in general, the current world operate.

Electricity demand also plays an important role in the amount of electricity that must be generated at any given time. Electricity demand is influenced by many factors such as time of day and

seasons, which are natural and everyday occurrences, but relatively rare factors such as the COVID-19 lockdown that have had a huge impact.

Therefore, it would be interesting to see how electricity demand influences this relationship, using the dataset from COVID-19 time-period (2019-2021) which is one of the most monumental events that has happened in the 21st century. During this time, large-scale societal changes such as lockdowns and reduced economic activity affected electricity consumption patterns.

Overall, this provides insight into how the fossil-renewable relationship responds to changing conditions and external factors.

Previous work Carbon Brief analysis shows that renewable electricity generation in the UK increased from 23 TWh in 2008 to 135 TWh in 2023, while fossil fuel generation fell from 303 TWh to 104 TWh. Fossil fuels generated 76 percent of UK electricity in 2008; since then, their share has fallen to a record low of 33 percent in 2023 as renewable energy exceeded. [2]

A study by Jiang et al. (2021) found that reduced electricity demand during the COVID-19 pandemic had a greater impact on fossil fuel generation, noting that “the reduced electricity demand had an acute impact on the generation and supply of fossil fuels like coal rather than those cheaper renewables.” [3]

A study by Liu et al. (2020) found that during the COVID-19 pandemic global energy demand decreased by 1.3 percent, leading to a 6.1 percent reduction in fossil-fuel electricity generation, while low-carbon sources increased by 6.2 percent, with renewable and low-carbon energy overtaking fossil-fuel sources unprecedentedly in May 2020. [4]

Objectives The aim of this project is to investigate the relationship between renewable electricity generation and fossil fuel electricity production in the UK power grid. In particular, the project explores how electricity demand influences this relationship and whether generation patterns changed during the COVID-19 period.

3 Data

Data provenance The data was obtained by the GridWatch, which provides historical and near real-time data on electricity generation and demand in the Great Britain power grid[5]. Gridwatch collects the data from original sources: Sheffield Solar at the University of Sheffield, which provides estimates of solar generation [6], and Elexon BM Reports system[7]. Gridwatch scrapes data from the Elexon Balancing Mechanism Reporting Service (BMRS), commonly referred to as BM Reports, at 5-minute intervals and stores it in a database, allowing users to do statistical analysis. The data is publicly available and can be downloaded as CSV files, allowing it to be used for academic analysis in this project.

Data description Data was arranged as a .csv file containing electricity generation and demand records from the Great Britain power grid. The dataset covers the period from January 2019 to December 2021. The data is recorded at five-minute intervals, resulting in 315,420 observations over the time period. The variables ID and nuclear were not used in the analysis, but the following variables were used in the analysis:

- timestamp contains the date and time of each observation.
- demand represents the total electricity demand on the grid measured in megawatts (MW).
- coal, ccgt, wind, solar, hydro, biomass represent electricity generation from different energy sources, also measured in megawatts (MW).

Data processing The dataset was processed in Python using the pandas library. Column names were cleaned by removing any trailing spaces to ensure consistent variable naming. The timestamp variable was converted from a string format into a datetime object to allow time-based analysis. The dataset was checked for missing values, and no missing observations were found in any variables. The variables ID and nuclear were removed as they were not required for this study. Two new variables were created to simplify the analysis: renewables, representing the combined electricity generation from wind, solar, hydro, and biomass, and fossil fuels, representing the combined generation from coal and combined cycle gas turbines (CCGT). Finally, observations where fossil fuel generation was equal to zero were removed.

These corresponded to periods where demand was below 10,000 MW and renewable generation was also zero, to avoid unrealistic values affecting the analysis.

4 Exploration and analysis

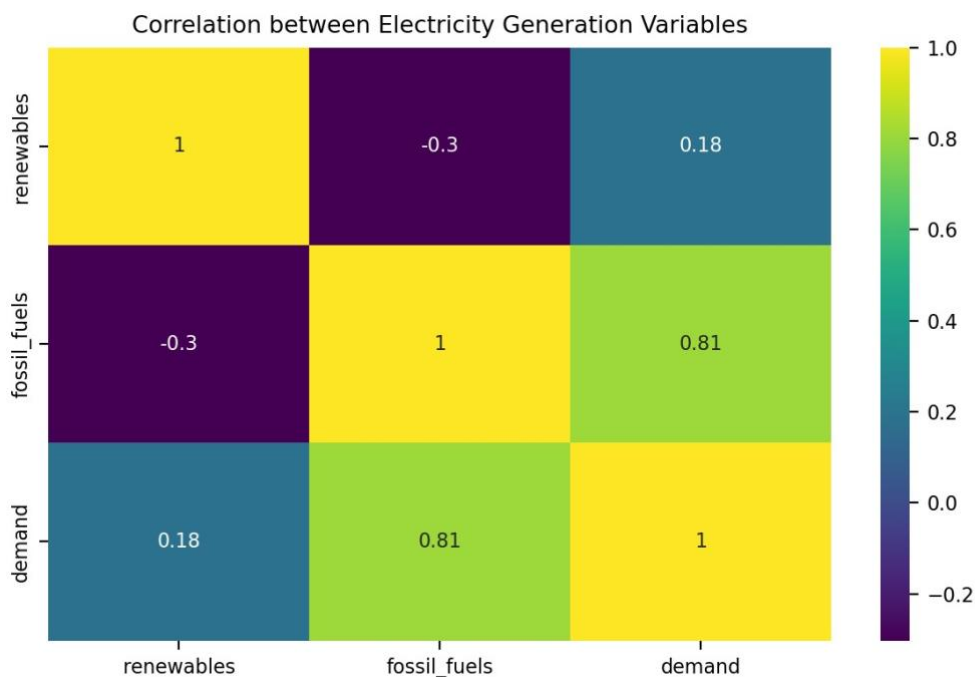


Figure 1: Pearson correlation coefficients between renewable generation, fossil fuel generation, and electricity demand.

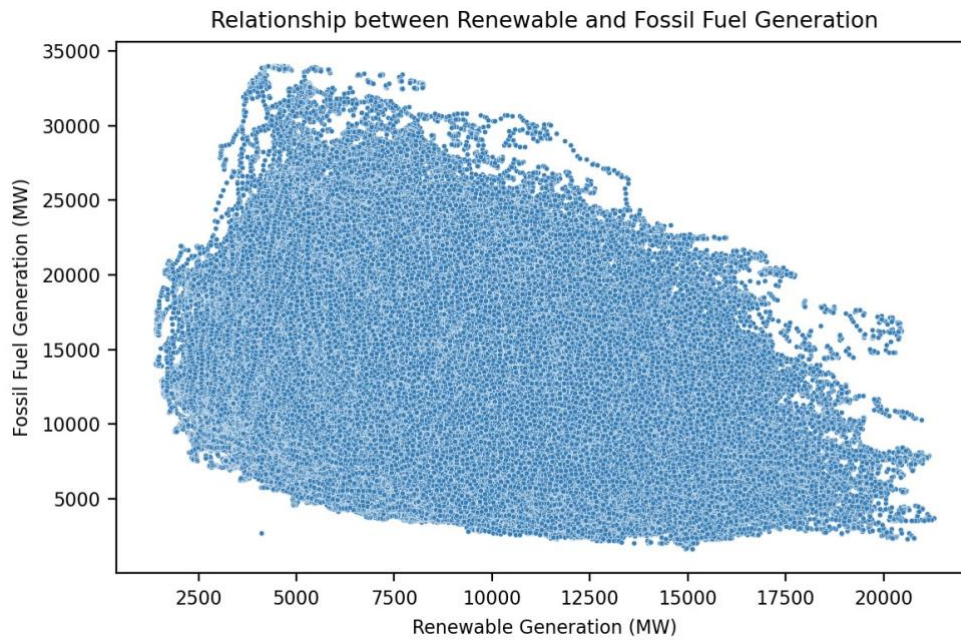


Figure 2: Scatter plot showing renewable electricity generation versus fossil fuel generation in the Great Britain power grid from 2019 to 2021. Each point represents a five-minute observation.

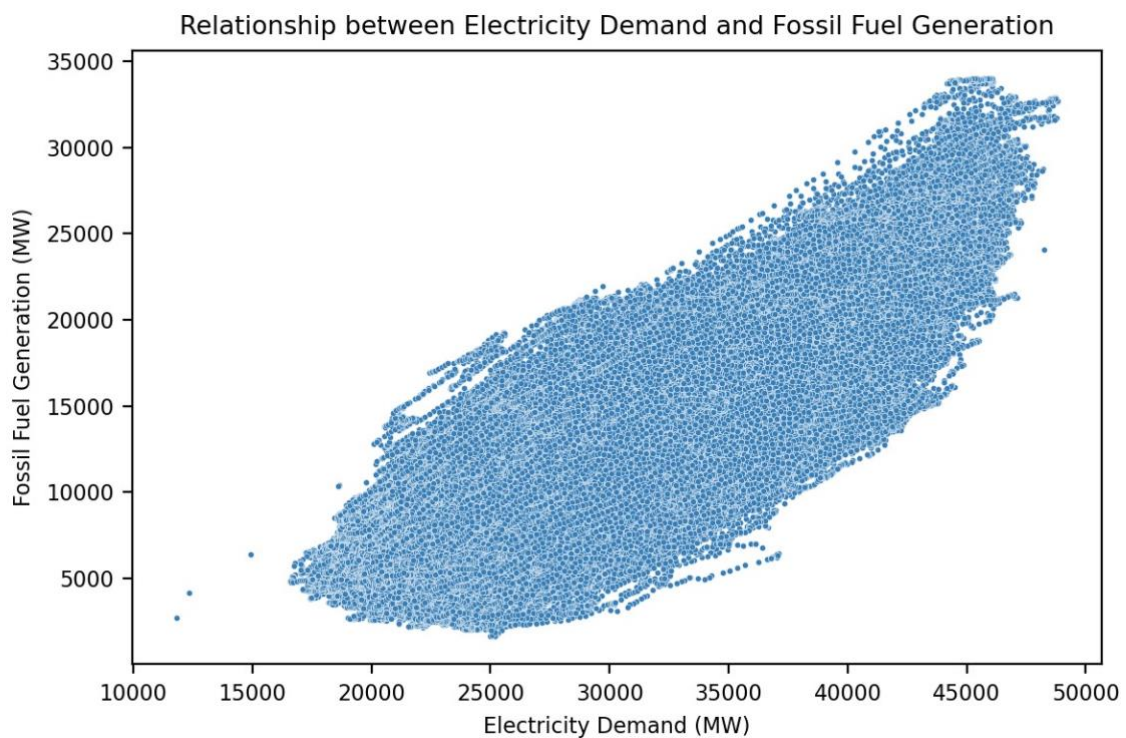


Figure 3: Scatter plot of electricity demand versus fossil fuel generation in the Great Britain power grid (2019–2021). Each point represents a five-minute observation.

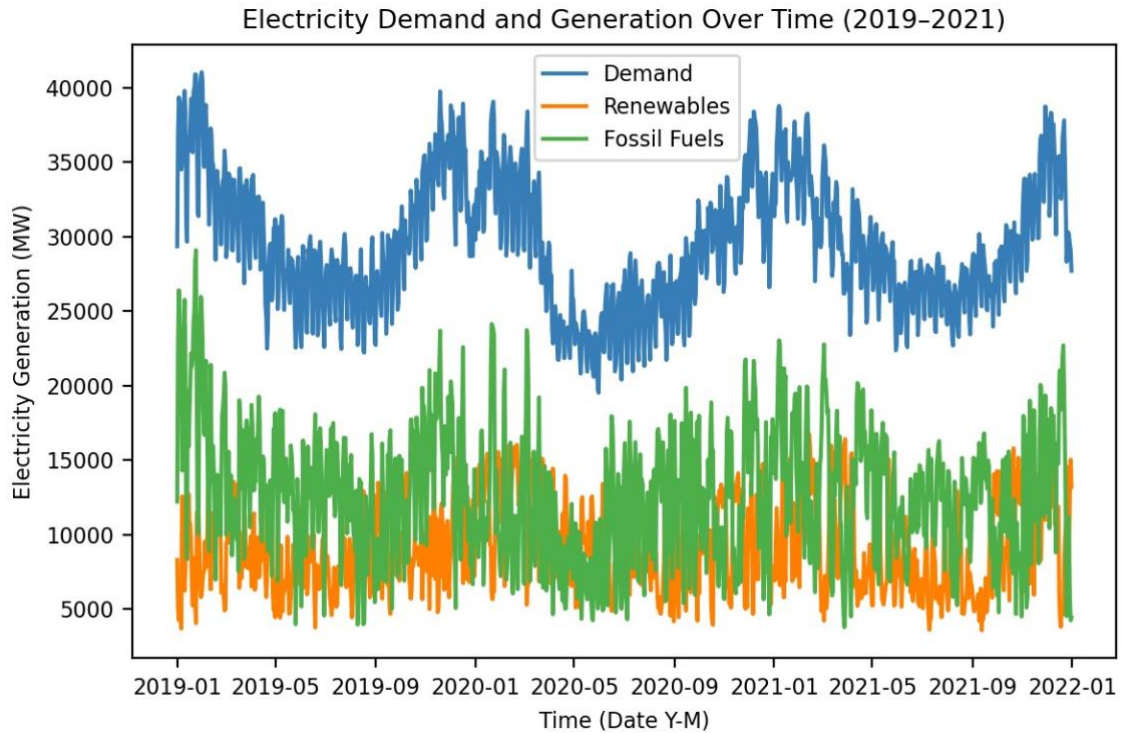


Figure 4: Time series of electricity demand, renewable generation, and fossil fuel generation in the Great Britain power grid from 2019 to 2021. Each observation represents a five-minute measurement. Interpretation of findings Figure 1 shows a moderate negative correlation between renewable and fossil fuel generation (-0.30), suggesting that increases in renewable generation are generally associated with reductions in fossil fuel generation. In contrast, fossil fuel generation has a strong positive correlation with electricity demand (0.81), indicating that fossil fuels are often used to meet higher electricity demand. Renewable generation has only a weak positive correlation with demand (0.18), suggesting that renewable output is less directly driven by demand and more dependent on weather conditions such as wind and solar availability.

Figure 2 shows a generally negative relationship between the two variables where as renewable generation increases, fossil fuel generation decreases, potentially indicating that renewable electricity partially displaces fossil fuel generation. The wide band suggests fossil fuel generation can take a wide range of values for the same level of renewable generation, showing it's a weaker relationship and that stronger factors determine fossil fuel production, particularly electricity demand.

Figure 3 shows a clear positive relationship between the two variables where as electricity demand increases, fossil fuel generation also increases. The points create a relatively narrow diagonal band, indicating a strong relationship between demand and fossil fuel generation. This suggests that fossil fuel power plants are often meeting increases in electricity demand. However, while there is some variation in fossil fuel generation at similar demand levels, this variation is relatively small compared with the overall trend. Indicating that electricity demand is the main factor influencing fossil fuel generation, while other factors such as renewable generation having a smaller effect.

Figure 4 shows electricity demand, renewable generation, and fossil fuel generation in the Great Britain power grid between 2019 and 2021. Clear seasonal patterns can be observed, with electricity demand generally increasing during winter months as heating is needed, and decreasing during warmer periods. A sharp drop in electricity demand compared to other drops occurs around March-May 2020,

which is the initial COVID-19 lockdown period where economic activity and electricity consumption decreased [8]. The lowest demand falls to just below 20,000 MW during this period, while other drops generally reach at minimum, the mid-20,000 MW range. During this March-May 2020 period, fossil fuel generation also declined, reflecting the reduced demand for electricity. Renewable generation appears more variable throughout this period, as it depends mostly on weather conditions such as wind and solar availability rather than electricity demand alone.

Overall, these results suggest that while renewable generation can reduce fossil fuel use, electricity demand remains the dominant factor determining fossil fuel generation in the UK power grid.

Statistical Analysis To quantify the relationships observed in the visualisations, a multiple linear regression model (OLS) was used to examine how renewable generation and electricity demand influence fossil fuel generation.

$$\text{fossil_fuels} = -5214.20 - 0.7347(\text{renewables}) + 0.8373(\text{demand}) \quad (1)$$

Renewable coefficient is -0.7347, showing that when renewable generation increases by 1 MW, fossil fuel generation decreases by about 0.73 MW, holding demand constant. This suggests that renewables are associated with fossil fuel reduction. The Demand coefficient is +0.8373, showing that when demand increases by 1 MW, fossil fuel generation increases by about 0.84 MW, holding renewables constant. This shows fossil fuels respond strongly to demand increases. The p-values for both renewable generation and electricity demand were less than 0.001, indicating that both variables are statistically significant predictors of fossil fuel generation, suggesting that the relationships observed are unlikely to be due to random variation. The 95 percent confidence intervals for both coefficients do not include zero, further confirming that renewable generation and electricity demand significantly influence fossil fuel generation. The model has an R^2 value of 0.863, indicating that approximately 86.3 percent of the variation in fossil fuel generation is explained by the model.

5 Discussion and conclusions

Summary of findings Overall, the results show that increasing renewable electricity generation is associated with a reduction in fossil fuel electricity production in the UK power grid between 2019 and 2021. However, this relationship is strongly influenced by electricity demand. While renewable generation is largely determined by external factors such as weather conditions, fossil fuel generation responds more to electricity demand changes. During the COVID-19 lockdown period, a significant decrease in electricity demand led to a corresponding reduction in fossil fuel generation, while renewable generation remained relatively unaffected. This suggests that electricity demand is the primary driver of fossil fuel generation, with renewable energy partially displacing fossil fuel use.

Evaluation of own work: strengths and limitations One strength is the large sample size, which increases the reliability of results in a short period of time. This helps to identify anomalies and outliers, especially ones that may seem implausible but it is not e.g., solar generation being zero makes sense during night-time periods. This large sample size also reduces the risks of Type I and Type II errors. One strength is the statistical analysis and the multiple methods used, the model fit is strong, explaining 86.3 percent of variation in the fossil fuel production. Time-series, correlation matrix, regression, scatter plots, descriptive statistics all are techniques that provide comprehensive understanding of data from different angles, leading to more nuanced conclusions.

One weakness is that the data is observational, making it difficult to establish causality between variables. Other factors, such as weather conditions or electricity imports, were not included which may influence the results and limit the accuracy of the conclusions. One weakness is that the solar data is an estimation from Sheffield Solar, this can lead to small inaccuracies by overestimating or

underestimating the true value of a parameter. One weakness is the visualisation of the time-series data, where renewable generation is less clearly visible due to overlap with the fossil fuel variable. This makes it more difficult to compare trends between renewable and fossil fuel generation over time, therefore, may lead to mistaken conclusions. One weakness is that since the data is time-series, observations are likely linked over time rather than fully independent, which goes against the OLS assumption of independent errors and may make the results appear more statistically significant than they really are.

Comparison with any other related work The findings of this project are consistent with some of the previous work in this area. Carbon Brief has shown that fossil fuel generation has fallen as renewable generation has increased[2], although this covers a longer period of time, the conclusion is still consistent with the findings of this study that their relationship is a negative correlation, as shown by the correlation matrix and scatter plots.

Jiang et al. has also shown that the global electricity demand has decreased during COVID-19 which affected fossil fuel production rather than renewables[3], although this considers global energy demand, the conclusion is still consistent with the findings of this study where reduced demand during COVID-19 led to a decrease in fossil fuel generation while renewable generation remained relatively unaffected, as shown by time-series plot.

Improvements and extensions One improvement is to use additional variables such as weather data (e.g., wind speed, solar irradiance), electricity imports, and energy storage to explain variation in fossil fuel generation better. The time period could also be extended to analyse whether the relationships remain consistent or not over time, particularly up to May 2023 when COVID-19 ended as a global public health emergency. One improvement is to analyse renewable sources independently (e.g., wind and solar) to better understand their individual impacts on fossil fuel generation.

References

- [1] Kevin Richardson. *What is the Electric Power Grid, and How Does It Work?* SER Connect. Retrieved 12 March 2026. 2024. URL: <https://serconnect.co.uk/blog/what-is-the-electricpower-grid-and-how-does-it-work/>.
- [2] Simon Evans. *Analysis: UK electricity from fossil fuels drops to lowest level since 1957*. Carbon Brief. 2024. URL: <https://www.carbonbrief.org/analysis-uk-electricity-fromfossil-fuels-drops-to-lowest-level-since-1957/>.
- [3] Ping Jiang, Yanan Fan and Jiří Jaromír Klemeš. 'Impacts of COVID-19 on energy demand and carbon emissions'. In: *Renewable and Sustainable Energy Reviews* (2021). URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7834155/#b0115>.
- [4] Zhu Liu, Philippe Ciais, Steven J. Davis et al. 'De-carbonization of global energy use during the COVID-19 pandemic'. In: *Nature Communications* 11 (2020), p. 5172. URL: <https://arxiv.org/abs/2102.03240?>.
- [5] Gridwatch Templar. *About the Gridwatch Site*. Retrieved 15 March 2026. 2024. URL: <https://www.gridwatch.templar.co.uk/index.php>.
- [6] Sheffield Solar. *PV Live: Real-time estimates of GB solar photovoltaic generation*. University of Sheffield. Retrieved 15 March 2026. 2024. URL: <https://www.solar.sheffield.ac.uk/pvlive/>.
- [7] Elexon. *Latest news and industry updates*. Retrieved 15 March 2026. 2026. URL: <https://www.elexonportal.co.uk/news/latest>.

- [8] Zi Lin and Xiaolei Liu. 'Impact of Covid-19 pandemic on electricity demand in the UK based on multivariate time series forecasting with Bidirectional Long Short Term Memory'. In: *Energy* (2021).
URL: <https://www.sciencedirect.com/science/article/pii/S0360544221007040>.

FDS-Project-Code

March 25, 2026

Please note:

1. This document is a template for your project code notebook, and gives some hints, but is not an exemplar
2. Retain the 4 top level headings (“Imports”, “Data”, “Exploratory Data Analysis”, “Data Communication and Analysis”)
3. Edit, remove or insert other cells as you wish
4. Before submission, **restart and run the whole Notebook from start to finish**, so that the numbers by the code cells start at [1] and go up [2], [3], ...
5. Then save as PDF using **File→Save and Export Notebook as→PDF**

1 Imports

You can import any packages you would like, but make sure you know what they are doing.

```
[1]: import pandas as pd
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.formula.api as smf

# Colour blind colours
CB_color_cycle = ['#377eb8', '#ff7f00', '#4daf4a',
                  '#f781bf', '#a65628', '#984ea3',
                  '#999999', '#e41a1c', '#dede00']

sns.set_palette(CB_color_cycle)
mpl.rcParams['font.size'] = 8      # Set the font size to 8pt
mpl.rcParams['figure.dpi'] = 200  # Set the dots per inch to a reasonable
    ↪compromise between resolution and size
```

2 Data

Code to load and process the data.

```
[2]: # Load data
gridwatch = pd.read_csv('gridwatch.csv')

#Remove any accidental spaces at the beginning or end of column names
gridwatch.columns = gridwatch.columns.str.strip()

#Convert the 'timestamp' column from a string to a datetime object
gridwatch['timestamp'] = pd.to_datetime(gridwatch['timestamp'])

# Drop the 'id' column since it is not needed for the analysis
gridwatch = gridwatch.drop(columns=['id', 'nuclear'])

# Create new columns 'renewables' and 'fossil-fuels' that sum up their
↳corresponding sources
gridwatch['renewables'] = (
    gridwatch['wind'] +
    gridwatch['solar'] +
    gridwatch['hydro'] +
    gridwatch['biomass']
)
gridwatch['fossil_fuels'] = (
    gridwatch['ccgt'] +
    gridwatch['coal']
)

# Remove rows where fossil fuel generation is zero, leading to removing demands
↳< 10,000, leading to remove when 'renewables' =0
gridwatch = gridwatch[gridwatch['fossil_fuels'] > 0]
```

3 Exploratory Data Analysis

Exploratory Data Analysis of your chosen dataset, which should demonstrate that you have understood the structure of the data, identified anomalous or missing data, and used visualisation and descriptive statistics to explore possible relationships in the data.

```
[3]: gridwatch.head()
```

```
[3]:      timestamp demand coal ccgt wind hydro biomass solar \
0 2019-01-01 00:00:36  25691 0  6298 9347 424   1069  0.0
1 2019-01-01 00:05:36  25571 0  6185 9347 423   1068  0.0
2 2019-01-01 00:10:37  25361 0  5900 9300 419   1059  0.0
3 2019-01-01 00:15:37  25413 0  5931 9273 403   1061  0.0
4 2019-01-01 00:20:36  25322 0  5904 9219 403   1057  0.0

renewables fossil_fuels
```

```

0    10840.0    6298
1    10838.0    6185
2    10778.0    5900
3    10737.0    5931
4    10679.0    5904

```

```
[4]: gridwatch.info()
```

```

<class
'pandas.core.frame.DataFrame'>
Index: 315387 entries, 0 to
315419 Data columns (total 10
columns):
#    Column          Non-Null Count  Dtype
---  -
0    timestamp        315387        non-null
      datetime64[ns]
1    demand           315387 non-null int64
2    coal             315387 non-null int64
3    ccgt             315387 non-null int64
4    wind             315387 non-null int64
5    hydro            315387 non-null int64
6    biomass          315387 non-null int64
7    solar            315387 non-null float64
8    renewables       315387 non-null float64
9    fossil_fuels     315387 non-null int64
dtypes: datetime64[ns](1), float64(2),
int64(7) memory usage: 26.5 MB

```

```
[5]: gridwatch.describe()
```

```

[5]: timestamp    demand    coal \ count    315387 315387.000000
315387.000000 mean    2020-07-02 02:34:02.814149120
29576.829888    581.734368 min    2019-01-01 00:00:36
11814.000000    0.000000
25%    2019-10-01 20:08:04 24664.500000    0.000000
50%    2020-07-02 07:05:34 28810.000000    228.000000
75%    2021-04-02    33566.000000    890.000000
      03:18:10.500000
max    2021-12-31 23:25:37 48813.000000    7648.000000
std    NaN    6293.607243    892.193609

      ccgt    wind    hydro    biomass \
count 315387.000000 315387.000000 315387.000000 315387.000000
mean  12050.396262  5706.295979  423.504995  2061.195122
min    1620.000000   35.000000   0.000000  139.000000
25%    7637.000000  2639.500000  224.000000  1621.000000
50%    11666.000000 5203.000000  399.000000  2168.000000

```

```

75%    16144.000000    8634.000000    592.000000    2450.000000
max     27163.000000   14286.000000   1434.000000   3290.000000
std      5392.257907    3508.154312    241.201126    641.350207

```

```

              solar    renewables fossil_fuels
count          315387.000000          315387.000000
315387.000000 mean  1225.727048  9416.723145
12632.130630  min    0.000000   1401.000000
1620.000000   25%    0.000000   6454.006590
7872.000000
50%  6.240000  9256.000000  12072.000000  75%
1950.000000  12215.000000  16753.000000  max
9630.000000  21294.000000  34016.000000  std
1929.222931  3746.692816  5897.198065

```

```
[6]: (gridwatch['fossil_fuels'] == 0).sum()
```

```
[6]: 0
```

```
[7]: (gridwatch['demand'] < 10000).sum()
```

```
[7]: 0
```

```
[8]: (gridwatch['renewables'] == 0).sum()
```

```
[8]: 0
```

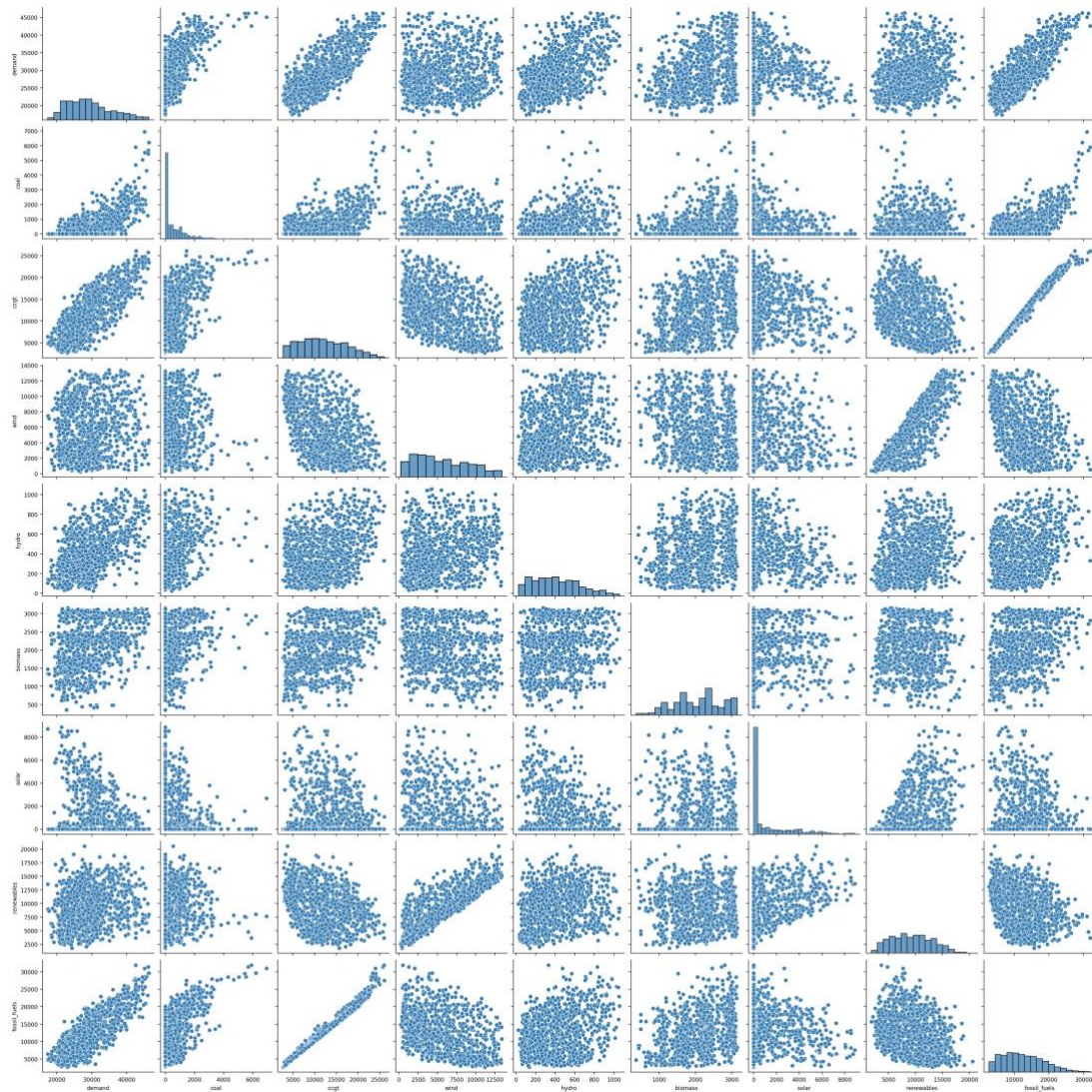
```
[9]: # No missing values
gridwatch.isnull().sum()
```

```
[9]: timestamp      0
demand             0
coal               0
ccgt               0
wind               0
hydro              0
biomass            0
solar 0 renewables
0 fossil_fuels    0
dtype: int64

```

```
[10]: sns.pairplot(gridwatch.sample(1000))
```

```
[10]: <seaborn.axisgrid.PairGrid at 0x7fe6b1d198e0>
```

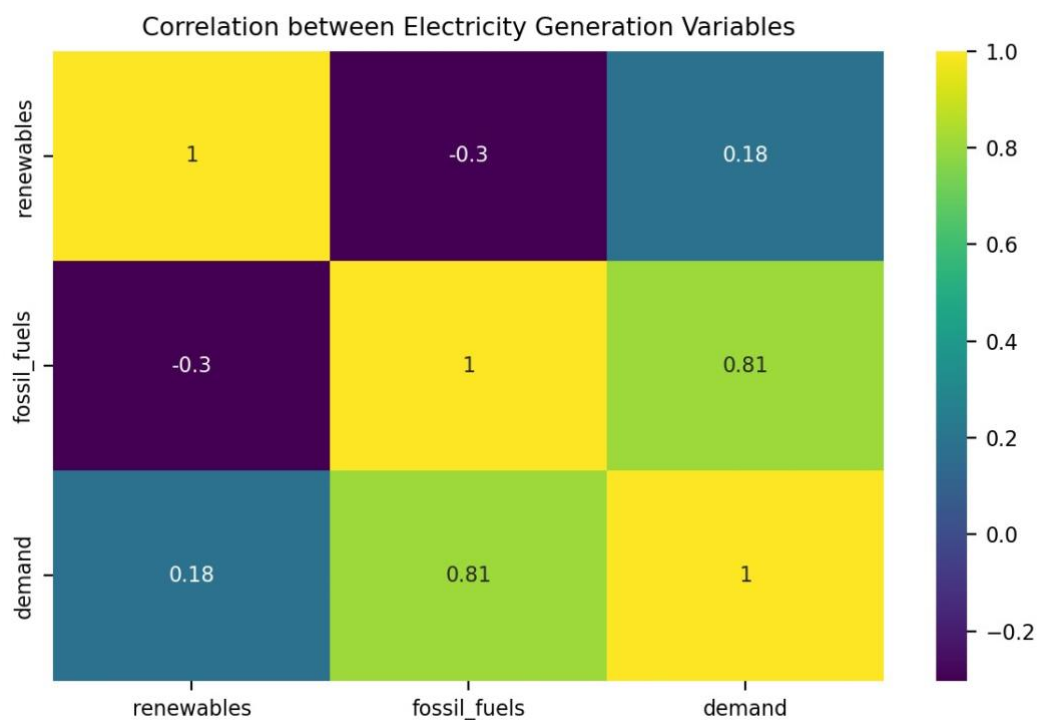


4 Data communication and analysis

The visualisation(s), tables and any statistical analyses or application of ML to the data, as presented in your report.

```
[11]: # Compute the correlation matrix for the selected variables
corr_matrix = gridwatch[['renewables', 'fossil_fuels', 'demand']].corr()
plt.figure(figsize=(6,4))

# Plot the correlation matrix as a heatmap, using colour-blind friendly
# 'viridis'
# annot=True displays the numerical correlation values inside each cell
sns.heatmap(corr_matrix, annot=True, cmap='viridis')
plt.title('Correlation between Electricity Generation Variables ')
plt.tight_layout()
plt.savefig('Corr_matrix.png')
plt.show()
```



```
[12]: plt.figure(figsize=(6,4))

# Create a scatter plot to show the relationship between renewable and
# fossil
# fuel generation, using s=5 points since dataset large
sns.scatterplot(
```

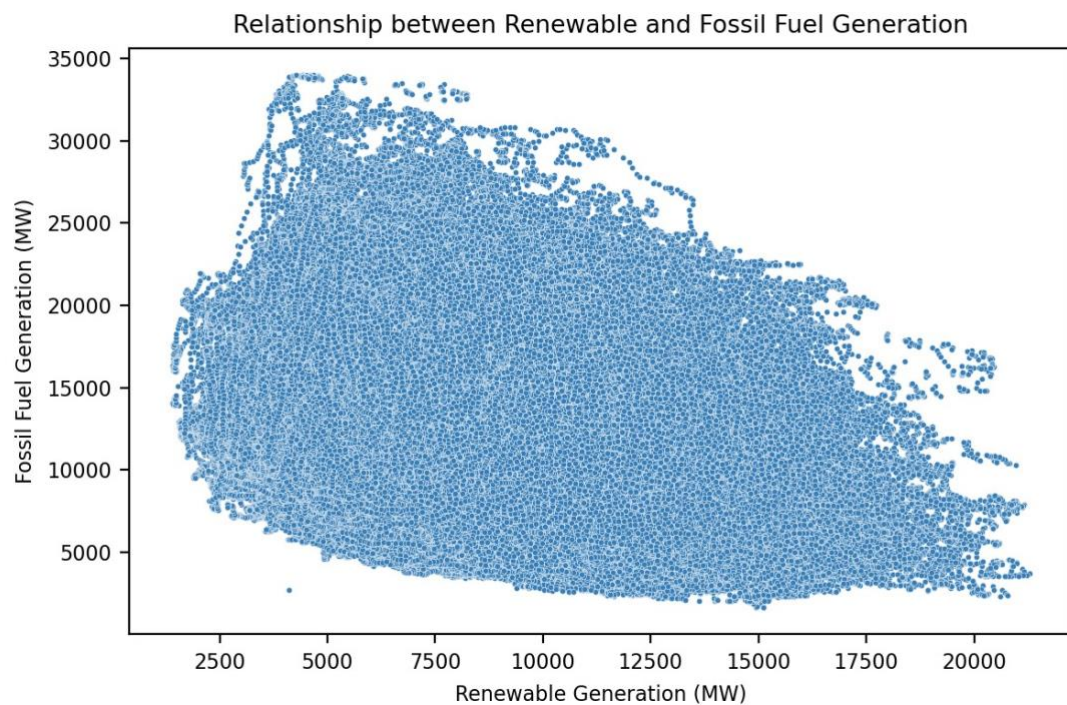
```

data=gridwatch,
x='renewables',
y='fossil_fuels',
s=5
)

plt.xlabel('Renewable Generation (MW) ')
plt.ylabel('Fossil Fuel Generation (MW) ')
plt.title('Relationship between Renewable and Fossil Fuel Generation ')

plt.tight_layout()
plt.savefig('Fossils_vs_renewables.png') # Save the figure as a PNG
plt.show()

```



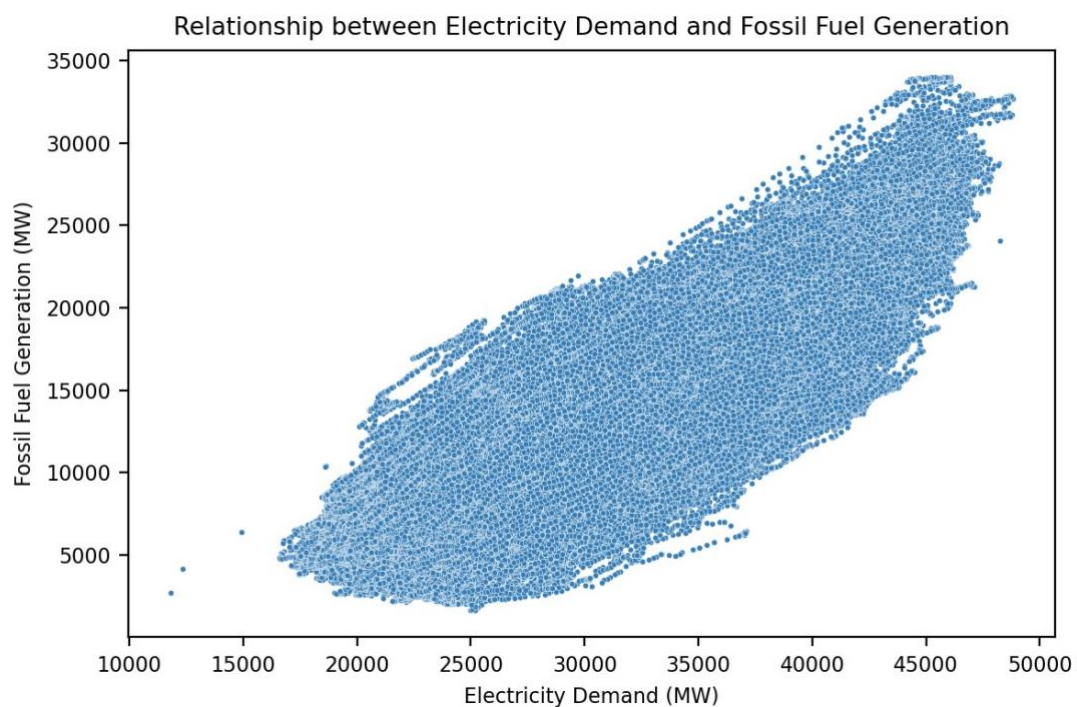
```
[13]: plt.figure(figsize=(6,4))

# Create a scatter plot to show the relationship between electricity demand and
# fossil fuel generation, using s=5 points since dataset large

sns.scatterplot(
    data=gridwatch,
    x='demand',
    y='fossil_fuels',
    s=5
)

plt.xlabel('Electricity Demand (MW) ')
plt.ylabel('Fossil Fuel Generation (MW) ')
plt.title('Relationship between Electricity Demand and Fossil Fuel Generation ')

plt.tight_layout()
plt.savefig('Fossils_vs_demand.png')
plt.show()
```



```
[14]: # Define an Ordinary Least Squares (OLS) regression model where
# fossil_fuels is the dependent variable,
```

```
# and renewables and demand are the independent variables used to
predict it model = smf.ols('fossil_fuels ~ renewables + demand',
data=gridwatch)

# Fit the regression model to the dataset, calculates the regression_
coefficients and other statistics

results = model.fit()

print(results.summary())
```

OLS Regression Results

```
=====
=====
Dep. Variable:          fossil_fuels R-squared:          0.863
Model:                  OLS Adj. R-squared:            0.863
Method:                 Least Squares F-statistic:      9.902e+05
Date:                   Wed, 25 Mar 2026 Prob (F-statistic): 0.00
Time:                   11:40:20 Log-Likelihood:        -
                                           2.8727e+06
No. Observations:      315387 AIC:                    5.746e+06
Df Residuals:          315384 BIC:                    5.746e+06
Df Model:               2
Covariance Type:       nonrobust
```

```
=====
===== coef      std err      t      P>|t| [0.025      0.975]
-----
-----
Intercept  -5214.1997   19.805   -263.274   0.000   -5253.017   -5175.382
renewables -0.7347  0.001  -695.124   0.000   -0.737   -0.733 demand  0.8373
0.001 1330.755  0.000  0.836  0.839
=====
=====
Omnibus: 22056.348 Durbin-Watson: 0.007 Prob(Omnibus): 0.000 Jarque-Bera
(JB): 26962.847
Skew:                0.697 Prob(JB):                0.00
Kurtosis:            3.326 Cond. No.                1.61e+05
=====
=====
```

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 1.61e+05. This might indicate that there are strong multicollinearity or other numerical problems.

```
[15]: # Set the timestamp column as the DataFrame index
gridwatch_time = gridwatch.set_index('timestamp')

# Resample the data by day ('D') and compute the daily mean
# Only the selected columns (demand, renewables, fossil_fuels) are
included daily_data =
gridwatch_time[['demand', 'renewables', 'fossil_fuels']].
↳resample('D').mean() plt.figure(figsize=(6,4))

# Plot daily electricity demand over time
sns.lineplot(data=daily_data, x=daily_data.index, y='demand',
label='Demand')

# Plot daily renewable energy generation over time
sns.lineplot(data=daily_data, x=daily_data.index, y='renewables',
↳label='Renewables')

# Plot daily fossil fuel generation over time
```

```

sns.lineplot(data=daily_data, x=daily_data.index, y='fossil_fuels',
             label='Fossil Fuels')

plt.xlabel('Time (Date Y-M) ')
plt.ylabel('Electricity Generation (MW) ')
plt.title('Electricity Demand and Generation Over Time (2019-2021) ')

plt.legend(loc='upper center', bbox_to_anchor=(0.513, 1))
plt.tight_layout()

plt.savefig('generation_over_time.png')
plt.show()

```

